

SYSTEM

A **static system** is a [system](#) in which output at any instant of time depends on the input sample at the same time. In other words, the system in which output depends only on the present input at any instant of time then this system is known as the static system. A static system is a memoryless system.

A **dynamic system** is a system in which output at any instant of time depends on the input sample at the same time as well as at other times. In other words, the system in which output depends on the past and/or future input at any instant of time then this system is known as the dynamic system. A dynamic system possesses memory.

System Modelling: System modeling is the process of developing abstract models of a system, with each model presenting a different view or perspective of that system. System modelling has generally come to mean representing the system using some kind of graphical notation, which is now almost always based on notations in the Unified Modeling Language (UML). However, it is also possible to develop formal (mathematical) models of a system, usually as a detailed system specification. A system [model](#) represents aspects of a system and its environment

A [model](#) is a simplified representation of a system at some particular point in time or space intended to promote understanding of the real system. As an abstraction of a [system](#), it offers insight about one or more of the system's aspects, such as its [function](#), [structure](#), properties, performance, [behavior](#), or [cost](#).

What is Information Systems?

An information system is a set of interrelated components that collects, stores, analyses and disseminates information for effective business decision-making. For example, an organisation wishes to increase its profits by boosting its sales.

In such a case, an information system can help the organisation in making effective sales-related decisions by providing information on current market trends, needs and preferences of customers, availability of substitutes, etc.

Components of Information System

The following are the key components of information system:

- **Hardware:** It includes physical devices for information processing such as computers, keyboards, mouse, scanners, pen drives, etc.
- **Software:** It includes a set of instructions to direct the computer's hardware. Software is broadly divided into two types: programs and procedures. Programs comprise operating system and application software, whereas procedures include instructions for data entry and information distribution.
- **Data:** It includes facts and observations (collected from various sources) used by programs to produce meaningful information. Data is stored in a machine-readable form on disk or tape.

- **People:** They are one of the main components of an information system. People are responsible for the development and operation of an information system. They can be categorised into two types: specialists and end users.

Specialists include system analysts, software developers, system operators, programmers and other IT personnel. On the other hand, end users are individuals (market researchers, production managers, customers and accountants) who use information produced by a system for various purposes.

Types of Information Systems

Types of information system includes:

1. Transaction Processing Systems (TPS)
2. Management Information Systems (MIS)
3. Decision Support Systems (DSS)
4. Executive Information System (EIS)
5. Enterprise Resource Planning
6. Expert System

Transaction Processing System (TPS)

The daily operations of a business are tracked using this kind of information system. A Point of Sale (POS) system is an example of a transaction processing system. The daily sales are recorded using a POS system.

To keep track of the organization's daily business transactions, transaction processing systems are used. The operational management level users use them. A transaction processing system's primary goal is to provide routine responses to issues like;

- What is the outstanding due for Mr. Ali?
- How much stock do we currently have?
- How were printers sold today?

The TPS system quickly responds to the queries above by keeping track of daily company transactions.

Functions of Transaction Processing System

- The transaction processing system generates extremely detailed information.
- Operational managers make routine decisions that are well-structured.

Example of Transaction Processing System

For instance, banks that offer loans require that a person's employer have an MoU (memorandum of understanding) with the bank. The only thing the operational staff needs to do when a borrower whose employer has an MoU with the bank applies for a loan is to check the given paperwork.

The documents for the loan application are processed if they satisfy the standards. The client is recommended to meet with tactical management staff to discuss the option of signing an MoU if they do not meet the standards.

Other examples of transaction processing systems include;

- Flight booking management using **flight booking systems**
- **Payroll systems**: managing loans, processing employee salaries, etc.
- **Stock Control systems** – monitoring stock levels
- **Point of Sale Systems** – records daily sales

Decision Support Systems (DSS)

Top level managers use decision support tools to make semi-structured decisions. The decision support system receives its output from the management information system as input. DSS systems also get data input from external sources like competitors, current market forces, etc.

Senior management utilises decision support systems while making non-routine decisions. Both internal (such as transaction processing systems and management information systems) and external systems provide input to decision support systems.

Decision support systems' primary goal is to offer unique, ever-changing challenges with solutions. Decision-support systems provide answers to issues;

- How will a new competitor entering the market affect our sales?
- What effect will doubling the factory's production lot have on the performance of the workers?

Decision support systems are highly interactive and use complex mathematical models and statistical techniques (probability, predictive modelling, etc.) to deliver solutions.

Examples of Decision Support Systems includes Bank loan management systems: they are used to assess a loan applicant's creditworthiness and forecast the possibility that the loan will be repaid.

they give managers the ability to assess alternative approaches to achieving objectives. The goal is to identify the most effective method of accomplishing it.

For instance, the formula $\text{Total Sales} - (\text{Cost of Goods} + \text{Expenses})$ is used to determine a company's net profit. Senior executives will be able to make adjustments to the values for total sales, cost of products, etc. using a financial planning system to examine the impact of decisions on net profit and

determine the best course of action.

Management Information Systems (MIS)

Management Information System or 'MIS' is a planned system of collecting, storing, and disseminating data in the form of information needed to carry out the functions of management.

Tactic managers are guided to make semi-structured decisions using management information systems. The output of the transaction processing system serves as the input for the MIS system.

Tactical managers utilise management information systems (MIS) to track the organization's current performance status.

The tactical managers use the reports that the MIS system generates to monitor, regulate, and forecast future performance by routinely analysing the data with algorithms that aggregate, compare, and summarise the finding

According to Davis and Olson, MIS is an integrated user-machine system designed for providing information to support operational control, management control and decision making functions in an organisation. Thus, MIS can be defined as an information system that:

- comprises various tools for data analysis
- delivers information to support managerial functions, such as planning, organising, etc
- gathers information as per the defined rules

MIS is the process of recording, storing, and processing data to provide information that decision-makers can use to make daily decisions.

It involves the use of information technology, people, and business processes. Management Information Systems is the full name of the acronym MIS. The goal of MIS is to gather data from many sources and generate insights that promote company's growth.

Every business needs data to function. In the internet era, businesses are overloaded with data—and not just useful data. Data is gathered by businesses in the form of files, papers, cloud storage, hard copies, and soft copies. Data is primarily securely stored by the MIS.

A management information system (MIS) is used for data processing. Employees, supervisors, and staff have access to MIS within an organization. For daily tasks like printing invoices, paying bills, or conducting performance reviews, employees use MIS.

For instance: A bank keeps track of customer information, account deposits, and withdrawals.

Additionally, they employ MIS as a database for information to compare, examine, and save data. Hence, MIS aids businesses in making quick, correct judgements.

Employees receive initial training on the MIS interface, usage, and importance. Then, in certain businesses, training is provided to clients and customers who utilize the system rather than employees.

MIS focuses more on data analysis. An information system, at its core, is a shared framework that boosts the productivity of all firm employees

Example of Management Information Systems

For instance, data from a point-of-sale system can be utilized to examine trends in the sales of both successful and unsuccessful products. Future inventory orders can be made using this information, increasing orders for products that are performing well and decreasing orders for underperforming products.

Examples of management information systems include;

- **Human resource management system** – overall welfare of the employees, staff turnover, etc.
- **Budgeting systems** provide an overview of the amount of money spent both temporarily and permanently within the organization.
- **Sales management systems** – they get input from the point-of-sale system.

Tactical managers are responsible for the semi-structured decision. The tactical managers make judgement decisions based on their knowledge and the information provided by MIS systems to forecast how much inventory or products should be ordered for the second quarter based on the first quarter's sales.

Other Example of MIS

1- Imagine a big hospital with lots of employees and patients. Each department in the hospital employs nurses, doctors, and surgeons. Some work on a temporary basis, while others are permanent employees. Everything must be documented by the hospital's administration.

A health information system is installed by the hospital to keep track of every small detail. Every spending is tracked by the MIS as well. The result is less disruption and easy access to patient information for the doctors.

2- Let's say a multinational corporation has numerous domestic and international offices. There are 960 workers in total. It is not possible to keep a manual record of every employee.

As a result, the human resource department of businesses collects, record, and classify employee information using a human resource information system. The company uses MIS to track employees' absences and attendance. The business also decides to use a centralized database to hold a large amount of information.

Objective of MIS in Business organizations

The main objective behind using MIS is to help managers in making multifaceted business decisions with ease. Besides, an organisation implements MIS due to the following reasons:

- **To operate in the dynamic business environment:** The environment in which businesses operate is changing tremendously due to continuous advancement in technology. This leads to

an increase in the complexities of different business functions.

For instance, if an organisation has its branches in various countries, it faces difficulties in controlling operations performed at all branches. In order to function efficiently in such a complex business environment, MIS offers integrated information to all the branches.

- **To survive in competition:** By providing accurate and updated information about the latest changes and market trends, MIS helps an organisation to sustain its position in the market. With the help of updated information, managers can make various strategic decisions effectively.
- **To accomplish managerial functions efficiently:** In the current business scenario, managers are responsible for performing various functions such as business planning, decision making, and forecasting. MIS provides information in a presentable form that can be easily understood by managers, thereby helping them in performing their functions effectively and quickly. For example, MIS helps managers in comparing the sales of the last five years with the sales of the current year by representing sales data in a graphical or tabular form.

Challenges of MIS

What is MIS Challenges: There are three major **challenges of MIS**: high cost, training of employees and maintenance cost. These are briefly discussed below:

1. High Cost
2. Training of Employee
3. Maintenance Cost
4. Quality of output depends on input.

High Cost - Development of new computerized based information system is a problem for the organization due to the cost factor and it creates problems because with the change of time there is need of up-to-date of the information system.

Training of Employee - Employees should have the capacity of learning of the information system with the changing competitive and business environment; otherwise it will be difficult for the organization to stay in the market.

Maintenance Cost - Sometimes a problem arises due to server crash and website crash. Sometimes it leads to the loss of information. So, maintenance cost is needed to tackle the above problem.

Quality of output depends on input.

Limitations of MIS

Even though MIS has many benefits but it also has its limitations. **Limitations of MIS** are discussed below:

- dependence on data accuracy,

- cost of implementation,
- potential for data security breaches,
- lack of flexibility for rapid changes, and
- the need for skilled personnel.
- While MIS may solve some critical problems but it is **not a solution to all problems** of an organization.
- It **cannot meet the special demands** of each person.
- MIS if **designed in an improper manner** does not serve the management and hence is of little relevance.
- The MIS is **not good if the basic data is obsolete** and outdated.
- Mostly information provided by the MIS is in quantitative form. Hence, it **ignores the qualitative information** like the attitude of an employee

Advantage of MIS

- Generate Competitive Advantages
- Implementation of Management by Objectives Techniques
- Fast Reaction to Market Changes
- Change in Industry Structure
- Functional Use
- External and Internal Change
- Availability of Customer Data

Executive Information System (EIS)

It compiles reports and summaries that are simple to read. The display of data is also optimized by organizational systems, which reorganize data into graphs, spreadsheets, and statistical tables. Managers are able to process graphical data more quickly as a result.

Executive Information System (EIS) is a tool that supports decision making at the top level of an organisation. It presents information in the form of customised reports to help executive levels in making a comparison of trends. For instance, by providing a report on sales figures of an organisation in the past five years, EIS can help the top executives of the organisation in setting sales targets for the next three years. It presents a report in a graphical format that can be easily analysed by top-level executives in a short period of time.

In Figure, it is shown that. In 2008, the target sales level was higher than the actual sales volume. However, in 2009, the sales target declined considerably to motivate salespeople to make higher achievements. This information is useful for top-level managers to determine an optimum sales target for the succeeding year.

Advantages of EIS

As said earlier, EIS is basically designed for top-level executives to retrieve updated information in a graphical format so that they can understand the business scenario in less time.

The following are some important advantages of EIS:

- It enables top management to quickly access both internal and external information.
- It helps executive-level managers in handling strategic issues and making efficient decisions by providing updated information.
- It displays the relevant information in the form of graphics, tables, and equations, which helps top-level executives in making quick analysis.
- It establishes sync between executives working on different systems within an organisation.
- By providing correct and timely information, it helps executives in identifying critical success factors.

Disadvantages of EIS

Though EIS is highly competent in improving the strategic efficiency of an organisation, it is not free from faults. Some of the important disadvantages of EIS are as follows:

- Its functionality is limited as it is meant only for top-level management.
- Sometimes, the graphical representation may not answer the questions in the mind of top-level managers. This makes it difficult for top managers to perform complete analysis and arrive at a conclusion.
- In terms of functions, it is difficult to separate EIS from DSS as both offer similar decision-making tools.
- The organisations incur huge cost for its implementation.

Enterprise Resource Planning (ERP)

ERP, which stands for Enterprise Resource Planning, is a software program made to make daily operations of a business, from managerial to logistical, easier. It aids in maintaining balance with the essential corporate operations, such as order management, accounting, and human resource management. This program serves as a central hub for streamlining information flow and all organizational processes.

These are a few key ERP features:

- These programs are employed to manage an organization's resources.
- These web-based programs are accessible through all interfaces.
- ERP software is responsible of monitoring the organization's growth.
- All of the services required to run the business are integrated using this software.

Customer Relationship Management (CRM)

CRM, or customer relationship management, is a software tool that enables a business to communicate with current and potential customers. This customer relationship management (CRM) programme controls, automates, and coordinates sales. The goal of creating such an application is to foster and amuse prospect leads, which aids the company's growth in sales and overall performance.

The following are some key CRM features:

- It aids in improving sales performance.
- This is utilized to oversee and manage the organization's operations.
- All customer services are integrated into one location using this program
- Primarily focused on customer.

Importance of MIS in business

1- Assists in Data Management

MIS assists management in making complicated decisions by maintaining and managing important company data. The administration can easily obtain the important data whenever needed because it is arranged and stored in an organised manner.

2- Goal Setting

Setting goals is an important task for every firm, one that necessitates extensive study and growth. As the data in MIS reports is based on recent data analysis, it is deemed appropriate and important for establishing an enterprise's purpose. The analysis of the present market trend and the forecast of the next trend are also included in MIS reporting. So, it is challenging for any organization to ignore MIS and reporting.

3- Comparison Of Business Performance

The MIS database stores and maintains the company's relevant business data and information. As the MIS database is always accessible, it is possible to compare the firm's present performance to its accomplishments from the prior year in order to measure the organization's growth.

4- Helps in Strategic Planning

The company's strategic planning heavily relies on MIS reports. It assists in identifying the company's future requirements and in developing goals and a strategy based on those requirements. The

management information system report is useful in determining the resources needed to achieve the company's goal. Therefore, it is crucial that the data generated by MIS and reporting is accurate and trustworthy.

5- Trends Analyses

Management must set future objectives and create forecasts for strategic planning. So, accurate reports on current industry trends are crucial for developing such a plan. The current market trend is analysed by MIS, which then makes predictions about future trends based on this data.

6- Problems Identification

Information on every facet of company activity is provided in a MIS report. As a result, MIS reports are quite useful in locating the source of the problem in the event that management encounters a problem. Moreover, MIS and reporting are quite helpful in identifying a remedy to such a problem.

7- Increases Efficiency

The company's goals and strategy are developed using the relevant information provided by MIS and reporting. MIS reports can also be used to evaluate how well the company is performing. Thus, MIS is essential for boosting a company's productivity.

Characteristics of MIS

1- Future Oriented

In addition to exception-based reporting, MIS needs to consider the future. In other words, MIS shouldn't just provide information from the past or the recent past; rather, it should provide information based on projections from which actions may be taken.

2- Sub-System Concept

Due to the complexity of the MIS development process, one may frequently lose understanding. The system must be divided into manageable sub-systems that are more significant during the planning stage even though it is seen as a single entity.

3- Management Oriented.

The MIS must be designed using a top-down approach. According to the top-down approach, system development should begin with the identification of management requirements and overall company objectives.

The overall business plan should serve as the basis for the MIS development plan. The fact that MIS is management-oriented suggests that management actively directs system development efforts.

4- Long Term Planning

MIS is created over a sizable amount of time. A system like that does not emerge overnight. There is a significant planning component. Future business goals and requirements must be considered by the MIS designer.

5- Integrated

An essential quality of a management information system is integration. Integration is important since it can result in more insightful information.

For instance, balancing aspects like setup costs, workforce, overtime rates, production capacity, inventory level, capital requirements, and customer services is important to build an efficient production scheduling system.

6- Central Database

The interconnectedness of the functional systems is maintained through a central database. The master file with data pertaining to inventories, people, vendors, customers, etc. must be accessible to each system. It would make sense to collect data once, fully verify it, and then store it on a central storage device that any other subsystem could access.

7- System Approach A System's approach is used by the information system. The system's approach denotes a comprehensive approach to the study of the system and its performance in the context of the goal for which it was created.

8- Exception Based

The exception-based reporting principle, which states that an abnormal event, i.e., the maximum; minimum or predicted values fluctuate beyond tolerance limits, should guide the development of MIS. In such cases, the decision-maker should receive the necessary level of exception reporting.

9- Need-Based

Strategic planning, management control, and operational control levels should all be taken into consideration while designing and developing a MIS. In other words, MIS should be tailored to the unique requirements of managers at various levels of a organization's hierarchy.

Role of MIS

1. Decision Making
2. Coordination among dept
3. Comparison of performance
4. Finding out problems
5. Strategies

Benefits of MIS

An MIS's primary purpose is to gather data, either automatically or manually, and provide a report to aid management in making important business decisions. Using a MIS has a number of advantages, including the following:

- Better marketing and promotional decisions can result from excellent customer data management and analysis.

- MIS can give a business a competitive edge while assisting in reducing downtime for decision-making.
- Product development, operational effectiveness, and business-critical decision-making can all be enhanced using MIS.
- Organization and communication among employees can be facilitated by certain forms of MIS, such as office automation systems.
- Data and reports generated by MIS can help organizations see more clearly within their own structures.
- The identification of an organization's strengths and shortcomings through the use of MIS reports can result in the strategically reproducing of strengths and successes as well as the exercise of weaknesses and failures.

Importance of MIS

The importance of MIS is explained in the following points:

- It provides access to useful and timely information to managers.
- It lays a basis for logical decision-making.
- It distributes integrated information among different departments of an organisation.
- It establishes a rational basis for developing business strategies

Decision Support Systems

In the present business environment, decision making has become far more complicated than it was in the past. Moreover, if decisions are not made on time, organisations may not be able to exploit opportunities or defend against threats. The major problem that managers face while making decisions is the availability of abundant information.

Sometimes, abundance of information causes distortion of its accuracy, which makes it difficult for managers to make a complete analysis of the problem. This gives rise to a need for computerised decision support that can enable managers to make effective decisions on time.

Decision Support System (DSS) is an interactive information system that is developed to help managers in making critical business decisions by performing logical data analysis and testing hypothesis.

According to Hicks, DSS is an integrated set of computer tools that allow a decision-maker to interact directly with computers to create information useful in making decisions.

In the words of Keen and Scott-Morton, A DSS couples the intellectual resources of individuals with the capabilities of the computer to improve the quality of decisions. It is a computer-based support system for management decision-makers who deal with semi-structured problems.

DSS mainly gathers and presents information related to legacy and relational data sources, relative data figures, estimated figures based on assumptions and data available, and possible outcomes of different decision alternatives.

Characteristics of DSS

The main characteristics of DSS are explained as follows:

- It helps individuals in performing decision-making activities and processes.
- It is an interactive system designed to enable decision-makers or staff users to understand the procedure of interaction and operations performed.
- It is developed to be used as both routine and ad-hoc decision support task.
- It offers various tools to support multiple tasks related to decision making, such as data analysis, identification and design of alternatives, selection of alternatives and decision implementation.
- It is an independent system that gathers or duplicates data from other information systems or subsystems.

Expert system

An expert system is a computer system that emulates, or acts in all respects, with the decision-making capabilities of a human expert. Artificial Intelligence is a piece of software that simulates the behavior and judgment of a human or an organization that has experts in a particular domain is known as an expert system. It does by acquiring relevant knowledge from its knowledge base and interpreting it according to the user's problem. The data in the knowledge base is added by humans that are expert in a particular domain and this software is used by a non-expert user to acquire some information. It is widely used in many areas such as medical diagnosis, accounting, coding, games etc.